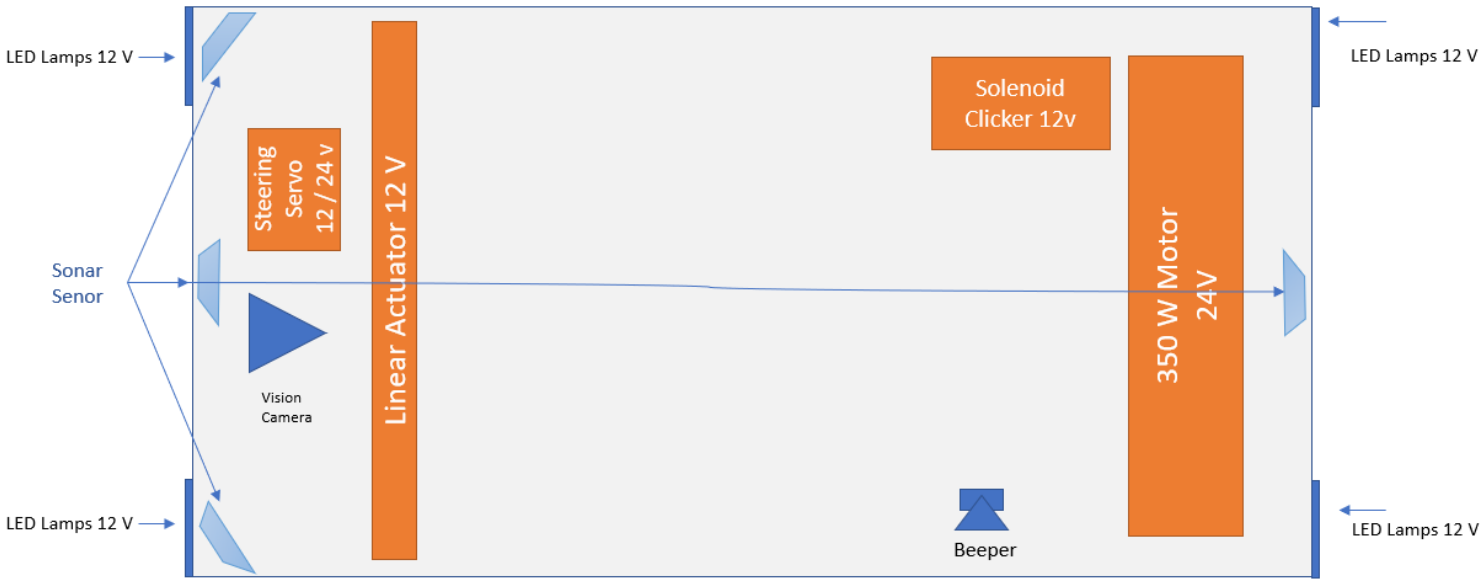
Scope of Work: Develop an Arduino based control board and sketches to run a RC cart wirelessly with following parameters:

1. **RC Communication 2.4 GHZ Remote**
2. **Camera System (FPV Camera)**
3. **Obstacle Detection (no avoidance functionality) only collision prevention**

Sketches will be for main Arduino board to control motors and a remote Arduino based controller system.

Carrier: The cart system that has the following components:

- LED Lamps
- Sonar Sensors
- Vision Camera
- Steering Servo
- Linear Actuator
- Solenoid
- Motor Transaxle
- Beeper



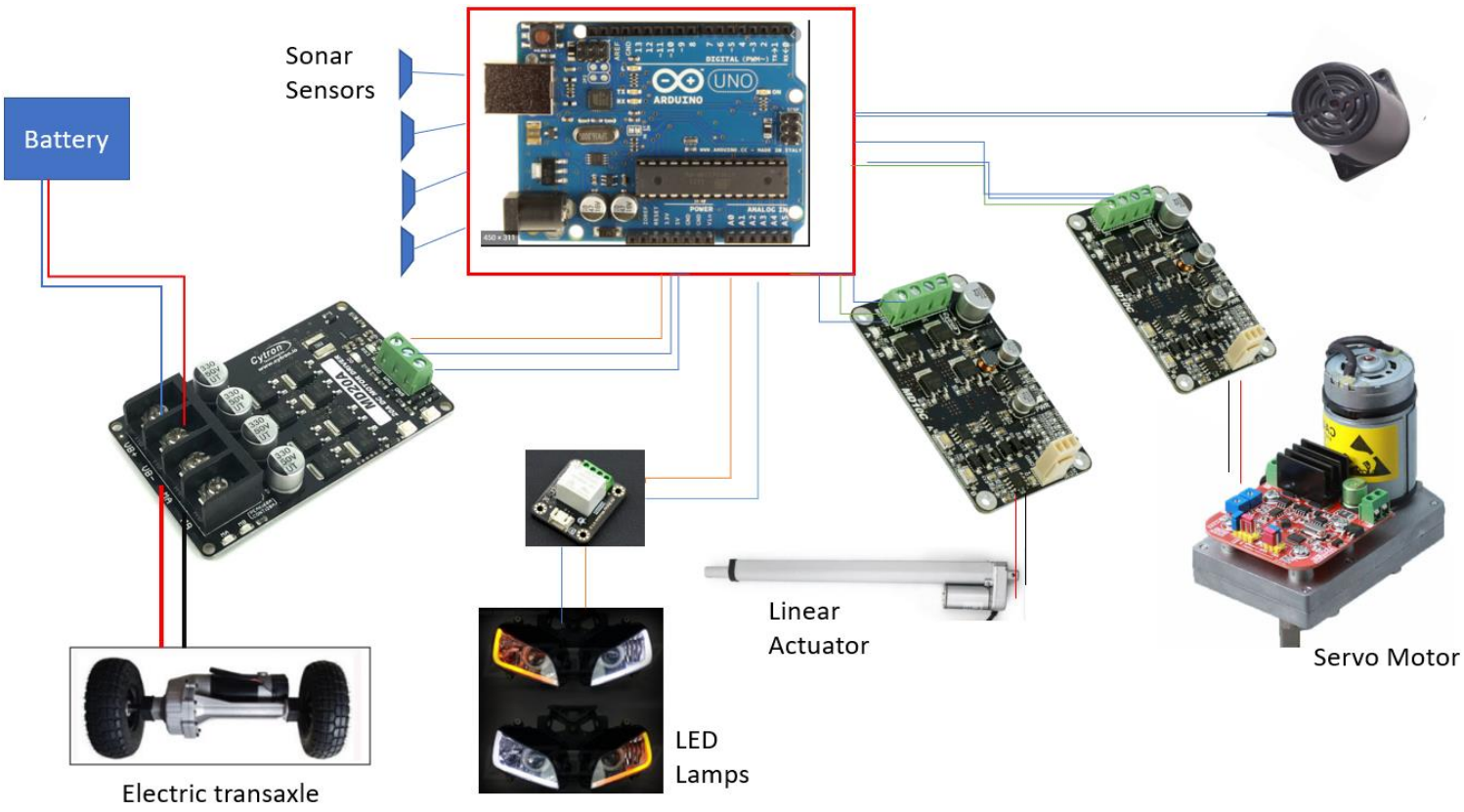
All Parts will be oulined in the Appendix Section and specs will be provided.

Task 1: Develop sketches for Arduino to control the following:

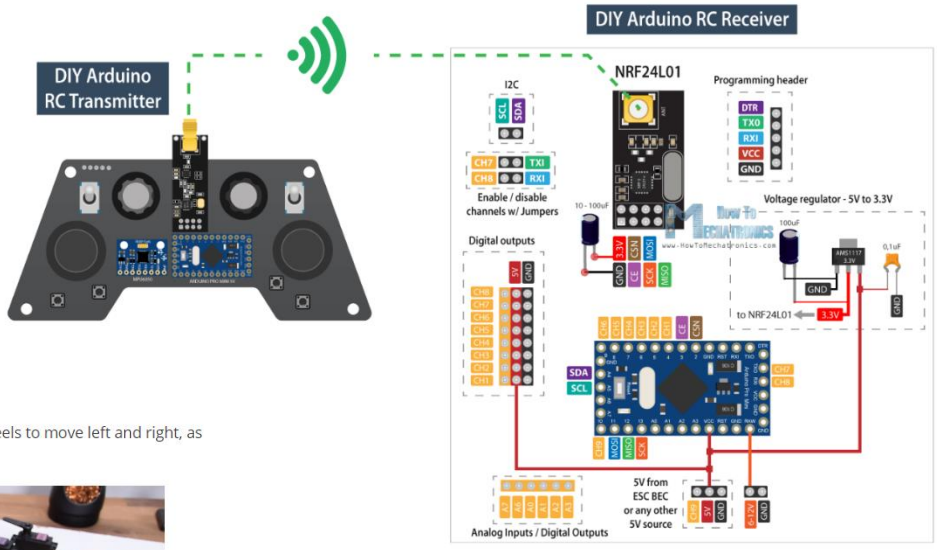
1. Power on the System
2. Drive 350 W brushed motor with motor controller
3. Steer the cart using servo motor with motor controller
4. Provide power feed to vision camera
5. Engage Sonar sensor to detect and perform collision avoidance
6. Open and close linear actuator based on remote command
7. Click electric soleniod using remote command.

The arduino should have a Keypad, to run these functions manually

The arduino should be also controlled by remote command.

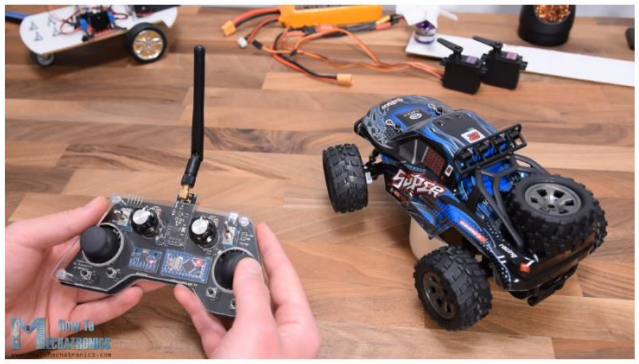


Task 2: Modify this remote with following functionality show in the Example 3:



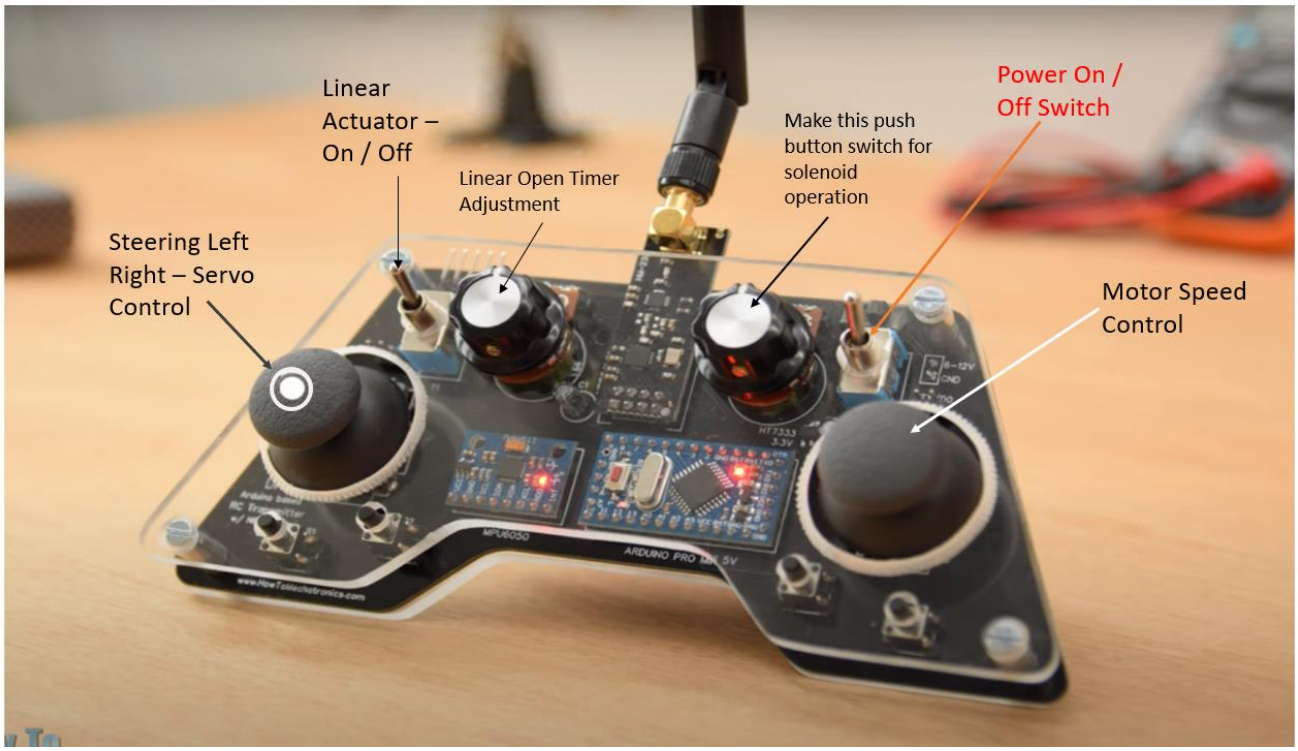
Example 3 – RC Car Model Control

The RC car came with its own controller which can control the front wheels to move left and right, as well as move the car forward and backward.



Link: https://howtomechatronics.com/projects/diy-arduino-rc-receiver/

Button Mapping:



Task 3: Provide instructions and guidance for uploading sketches, debugging any information and offer troubleshooting to make the arduino uno and remote work. Provide assitance with correct wiring and working of the units.

Detailed listing and specs for components will be shared once the pricing is agreed.

Deliverables:

1. Sketch for main Arduino Board
2. PCB board that plugs in on top of arduinio with connectors labelled to connect all devices on page 2
3. Sketch for Remote controller
4. PCB board and design for remote controller
5. Wiring diagram and sketch upload files and sequence for programming
6. Parts listing needed to assemble everything
7. Provide assistance and trouble shooting as needed to make the unit work
8. Offer Enhancements and Efficiency to design